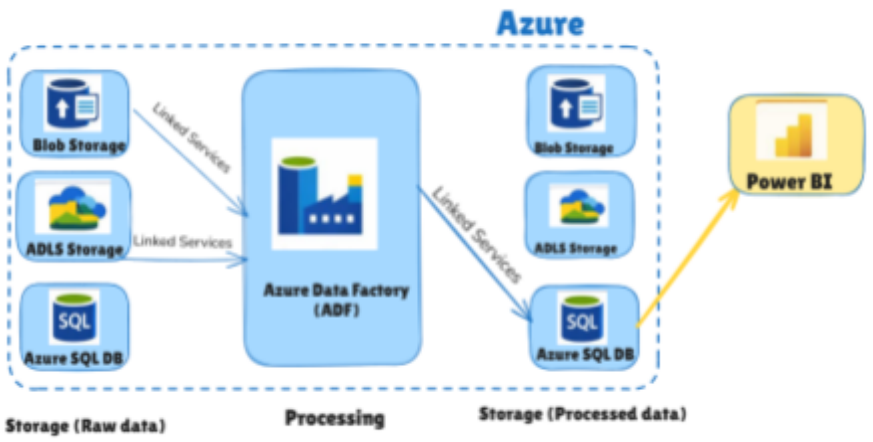
Azure Blob Storage + Azure Data Lake Storage to Azure SQL Database (Using Azure DataFlow)

Overview:

This document outlines the architecture, setup, and key considerations for moving data from Azure Blob Storage (ABS) and Azure Data Lake Storage (ADLS) into Azure SQL Database (SQL DB) using Azure Data Factory's (ADF) Mapping Data Flows. This approach is ideal for transforming and loading large volumes of heterogeneous data from low-code, scalable sources.

Project Architecture:



1. Architecture Overview:

The proposed solution utilises Azure Data Factory as the orchestrator and the Mapping Data Flow feature for complex data transformation and loading.

1.1 Components:

Component	Role	Source/Destination
Azure Blob Storage (ABS)	Stores unstructured/semi-structured source data (e.g., historical archives, raw logs).	Source
Azure Data Lake Storage (ADLS) Gen2	Stores large volumes of data, often optimised for analytical workloads and serving as a landing zone.	Source
Azure Data Factory (ADF)	Orchestration service for data movement and transformation. Contains Pipelines and Data Flows.	Orchestrator/ETL Tool
Mapping Data Flow (within ADF)	The visual transformation engine, where data shaping, cleansing, and aggregation occur.	Transformer
Azure SQL Database	Relational data store for structured data, serving applications and reporting layers.	Destination/Sink

1.2 Workflow:

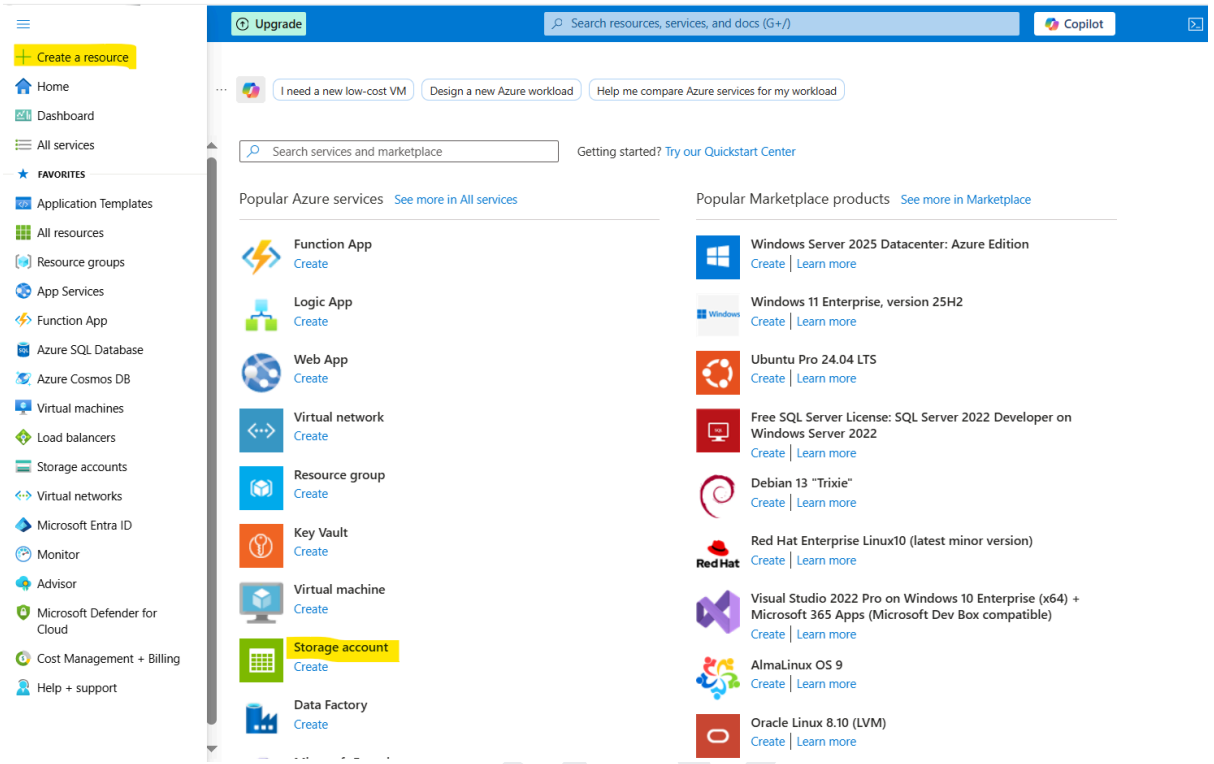
1. **Trigger:** An ADF Pipeline is triggered (e.g., schedule, event-based).
2. **Source:** The ADF Pipeline executes a Data Flow. The Data Flow sources data from registered Linked Services pointing to ABS and ADLS.
3. **Transform:** Data Flow applies necessary transformations (e.g., joins, lookups, column derivations, filtering, aggregation).
4. **Load:** The transformed data is written to the Sink, which is the Azure SQL Database Linked Service.

2. Setup and Configuration:

2.1 Azure Prerequisites:

- Azure Subscription
- Azure Blob Storage container(s) with source files

Dhanineni Babu


Upgrade
Search resources, services, and docs (G+/)
Copilot
Create a resource
Home
Dashboard
All services
FAVORITES
Application Templates
All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
Advisor
Microsoft Defender for Cloud
Cost Management + Billing
Help + support
I need a new low-cost VM
Design a new Azure workload
Help me compare Azure services for my workload
Search services and marketplace
Getting started? Try our Quickstart Center
Popular Azure services
See more in All services
Function App
Create
Logic App
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Web App
Create
Virtual network
Create
Resource group
Create
Key Vault
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Popular Marketplace products
See more in Marketplace
Windows Server 2025 Datacenter: Azure Edition
Create | Learn more
Windows 11 Enterprise, version 25H2
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Ubuntu Pro 24.04 LTS
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Free SQL Server License: SQL Server 2022 Developer on Windows Server 2022
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Debian 13 "Trixie"
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Red Hat Enterprise Linux10 (latest minor version)
Create | Learn more
Visual Studio 2022 Pro on Windows 10 Enterprise (x64) + Microsoft 365 Apps (Microsoft Dev Box compatible)
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AlmaLinux OS 9
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Oracle Linux 8.10 (LVM)
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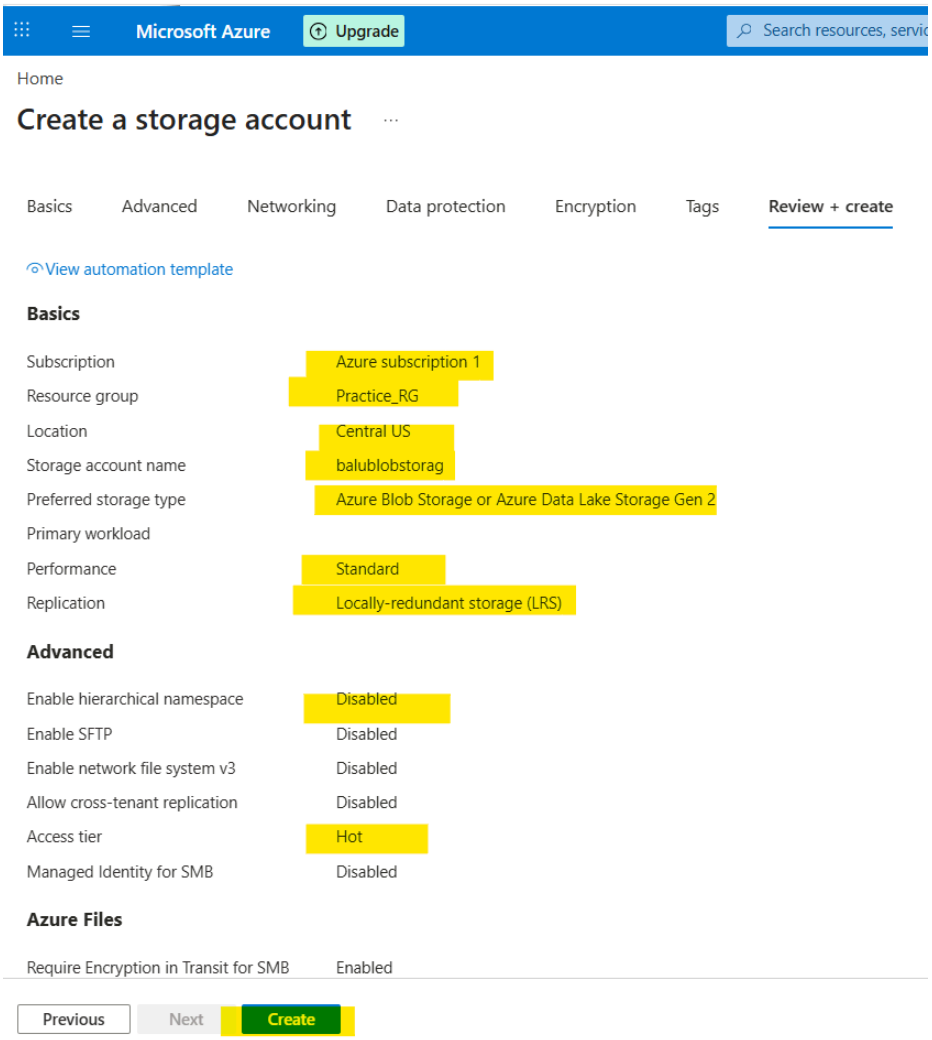
Microsoft Azure
Upgrade
Search resources, servic
Home
Create a storage account
Basics
Advanced
Networking
Data protection
Encryption
Tags
Review + create
View automation template
Basics
Subscription
Azure subscription 1
Resource group
Practice_RG
Location
Central US
Storage account name
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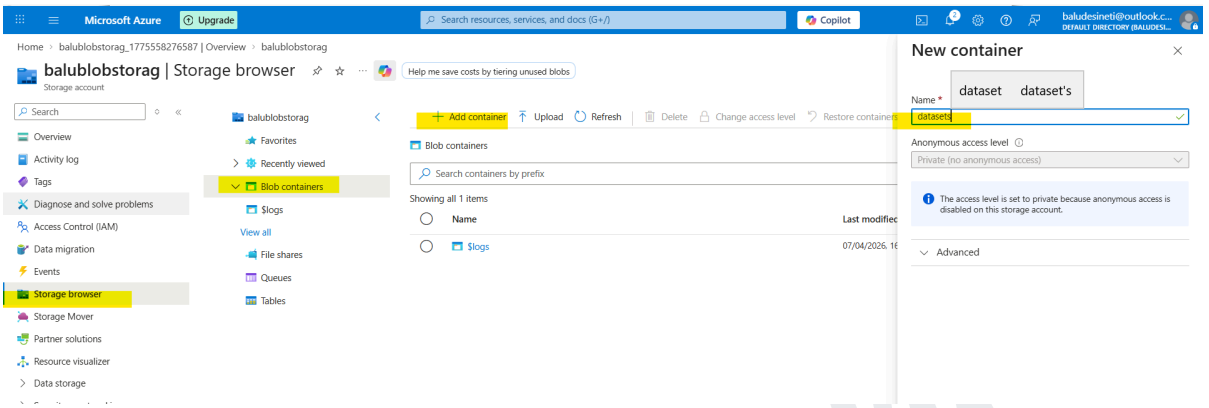
Preferred storage type
Azure Blob Storage or Azure Data Lake Storage Gen 2
Primary workload
Performance
Standard
Replication
Locally-redundant storage (LRS)
Advanced
Enable hierarchical namespace
Disabled
Enable SFTP
Disabled
Enable network file system v3
Disabled
Allow cross-tenant replication
Disabled
Access tier
Hot
Managed Identity for SMB
Disabled
Azure Files
Require Encryption in Transit for SMB
Enabled
Previous
Next
Create

Creating the step-by-step on Bolob Storage

Note: for blob storage, the hierarchical namespace should be disabled

Create the blob storage



After creating blob storage, go to the Resource group and navigate the

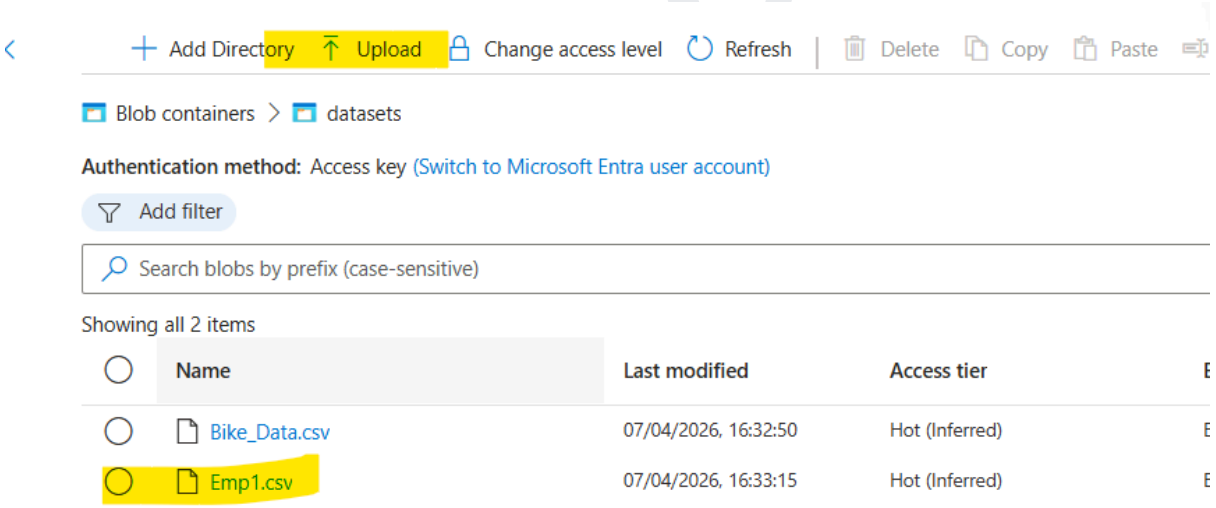
→ **Storage Browser**

→ **Blob Container**

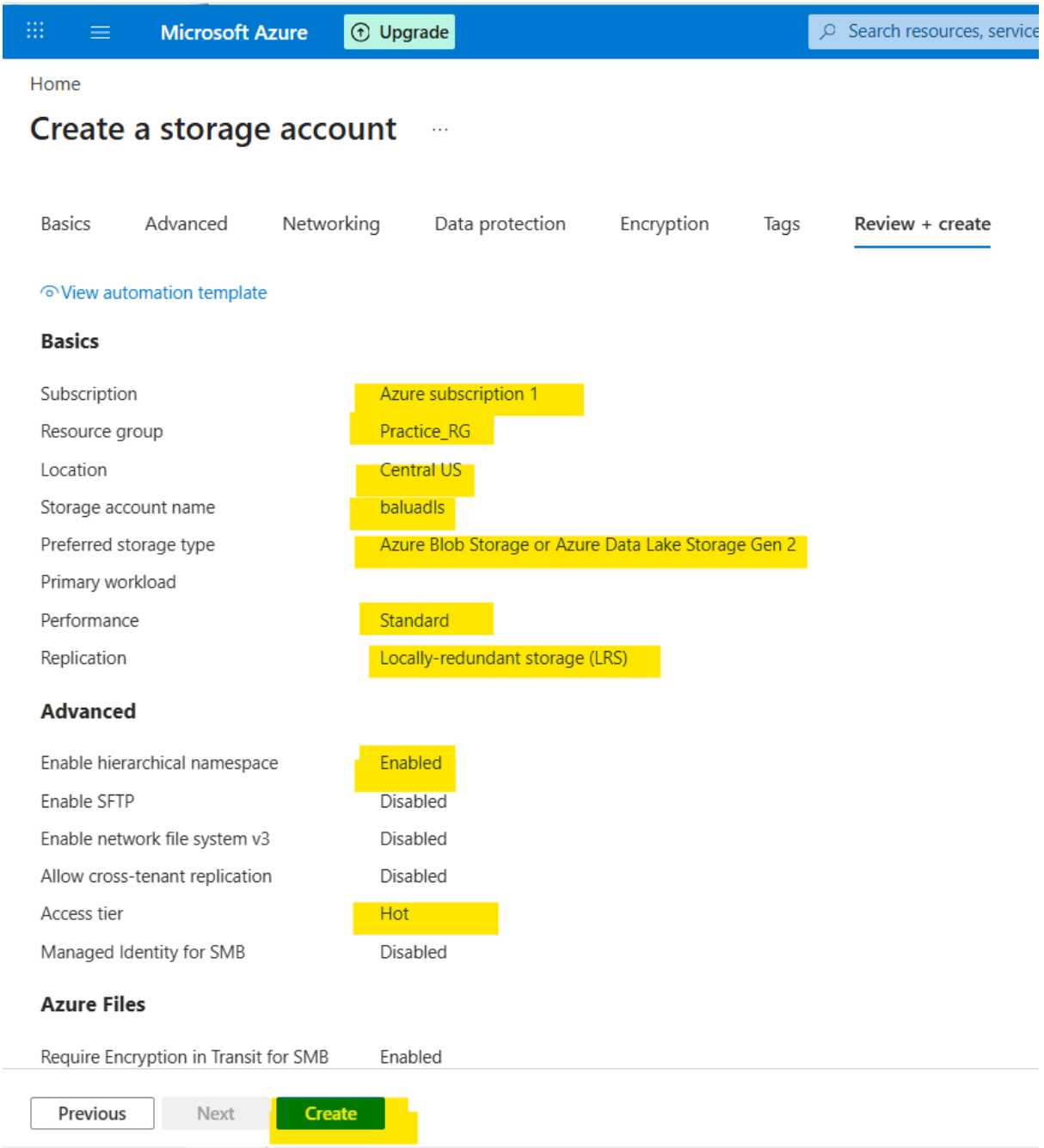
→ **Add Container**

→ **Name: ex, datasets and Create**

→ **Uploaded Emp Data**



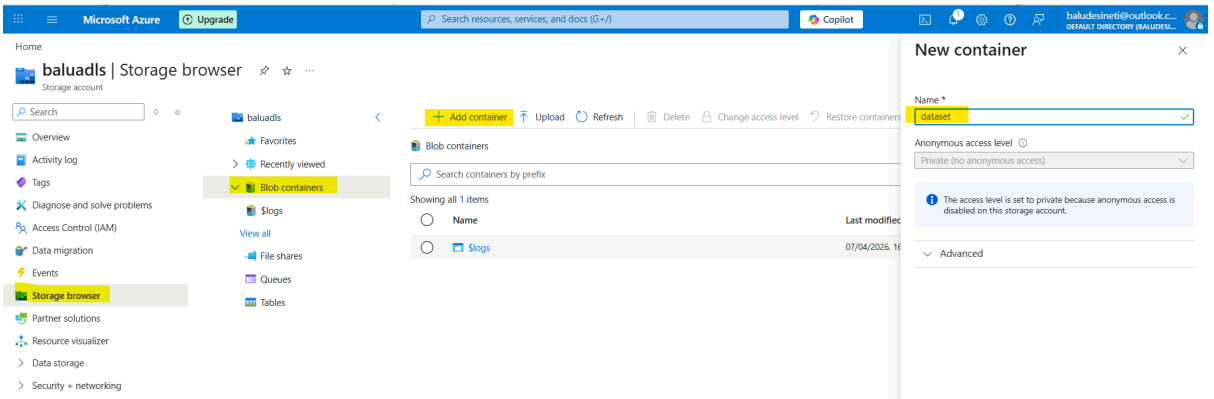
- Azure Data Lake Storage Gen2 account/filesystem with source files



Creating the step-by-step on Azure Data Lake Storage(ADLS)

Note: for blob storage, the hierarchical namespace should be enabled

Create the Azure Data Lake storage(ADLS)



After creating blob storage, go to the Resource group and navigate the

→ **Storage Browser**

→ **Blob Container**

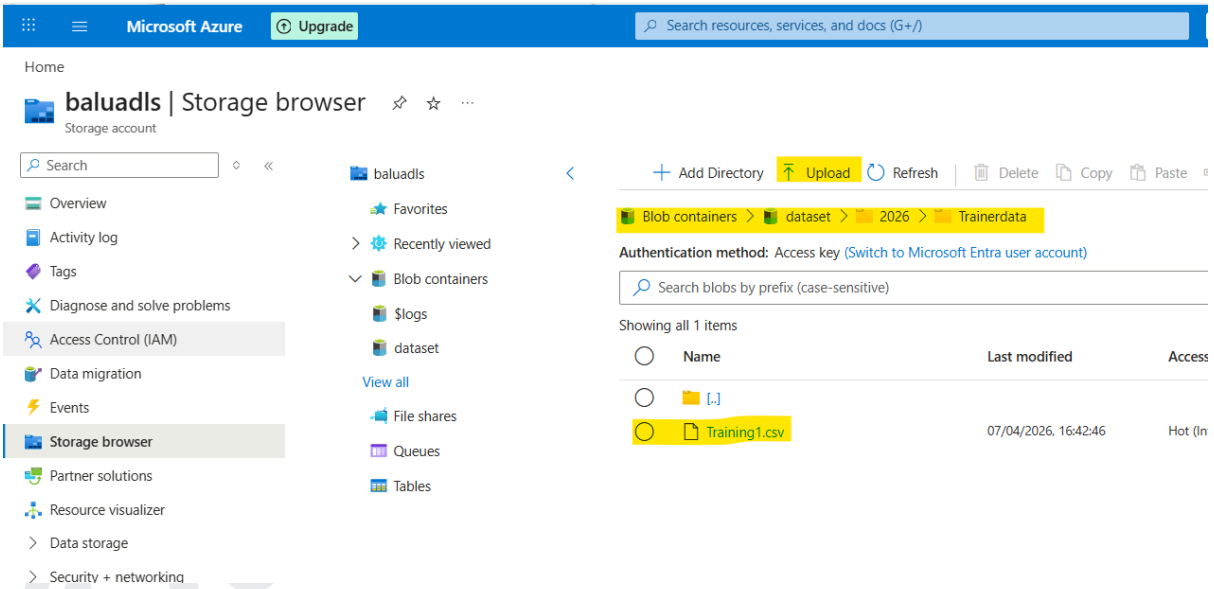
→ **Add Container**

→ **Name: ex, datasets and Create**

→ **Create Directory1 Name**

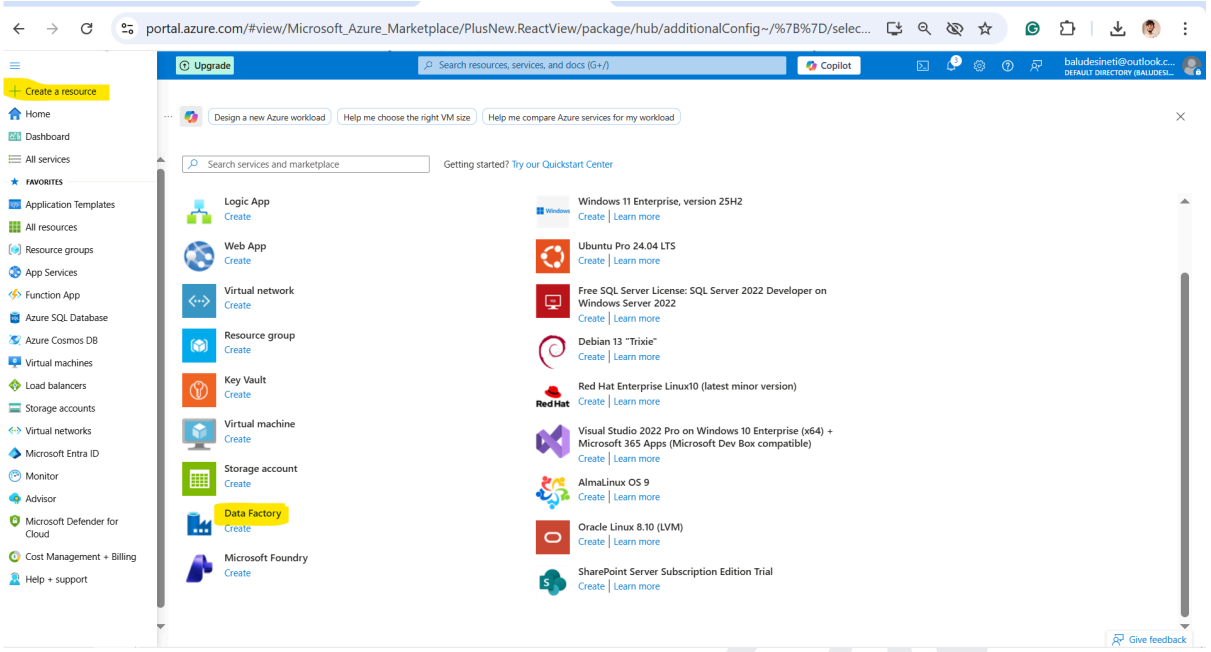
→ **Inside Directory: Create another Directory**

→ **Uploaded Trainer Data**



- Create Azure Data Factory

Navigate to Main Menu → go to the **Data Factory**

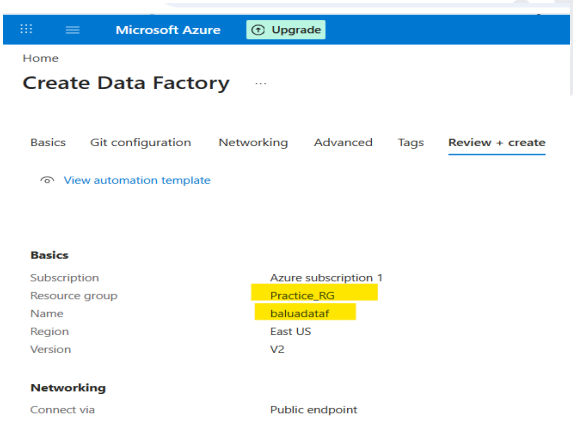


Basic Tab

Subscription: Azure Subscription 1 (default)
Region: Select Region(Ex, East US)
Version: V2

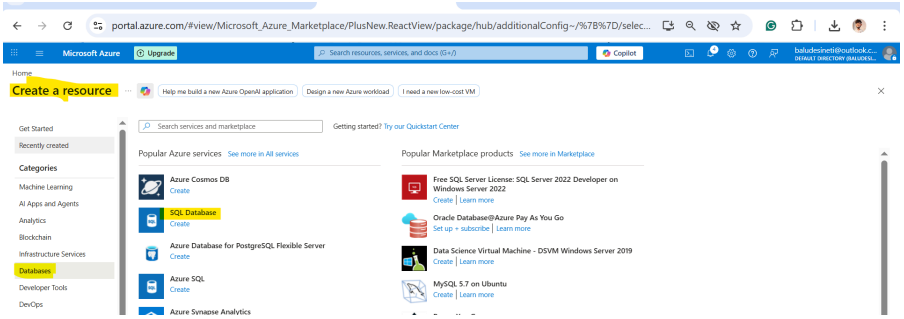
Networking Tab

Connect Via: Public endpoint



- Azure SQL Database instance and target table(s)
- Create Azure SQL Database

Navigate to Create Resource → Database → SQL Database.



Basic Tab

a) **Subscription:** Azure Subscription 1 (default)

b) **Resource group:** Create New (e.g., Practice_RG)

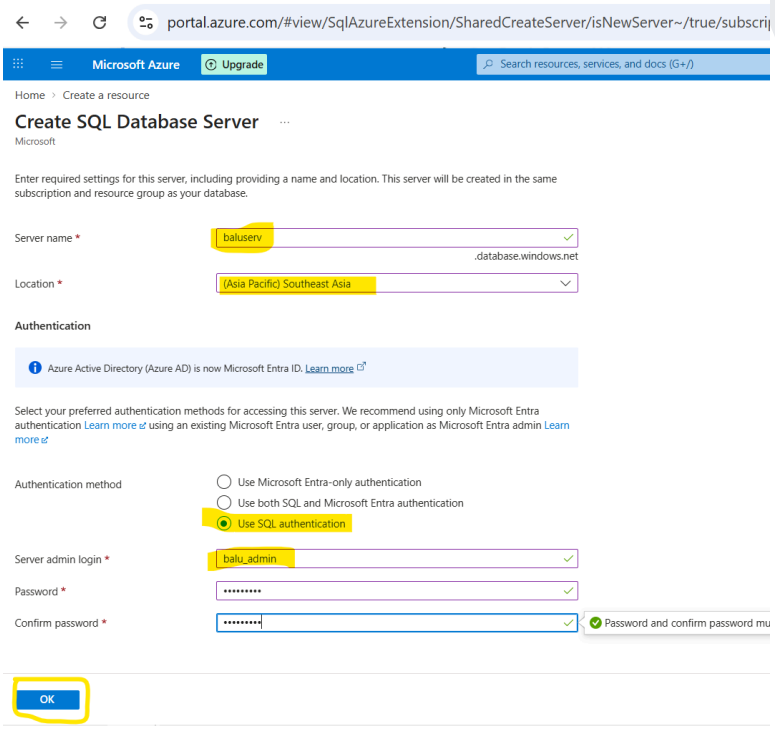
c) **Database Name:** give Database Name(baludb)

d) **Server:** creating server Name(baluserv)

e) **Location:** Asia Pacific Southeast Asia

f) **Authentication Method:** Use SQL Authentication

g) **server admin login:** admin and password



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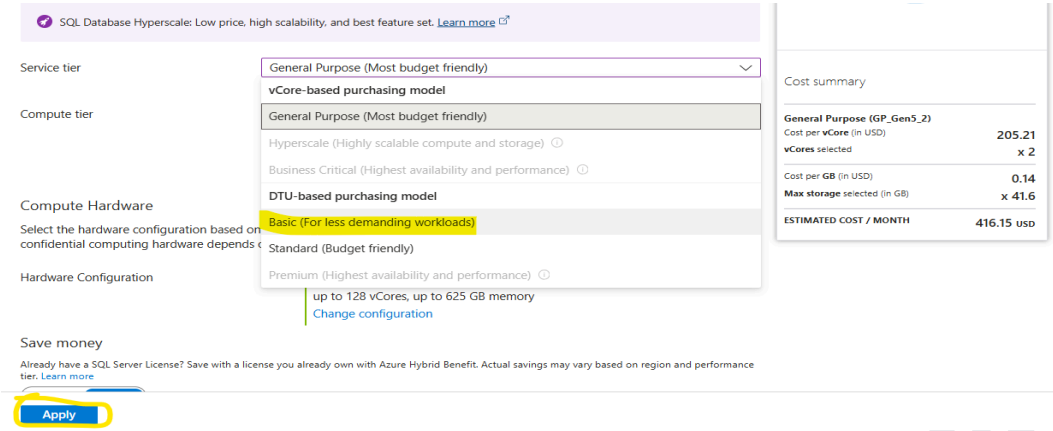
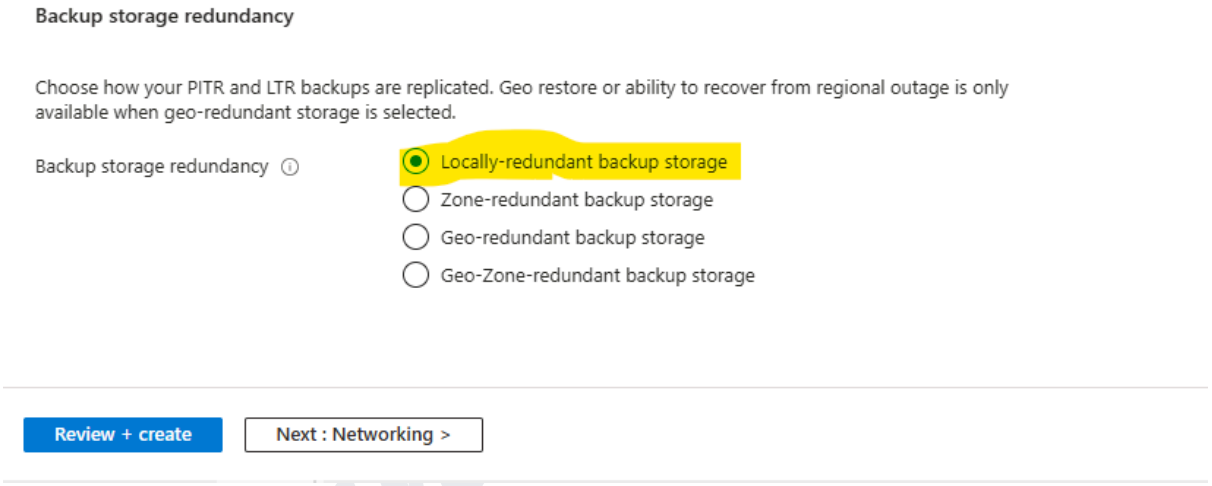
h)select core type: select DTU Basic core and apply



g) Backup Storage Redundancy: Locally-redundant backup Storage



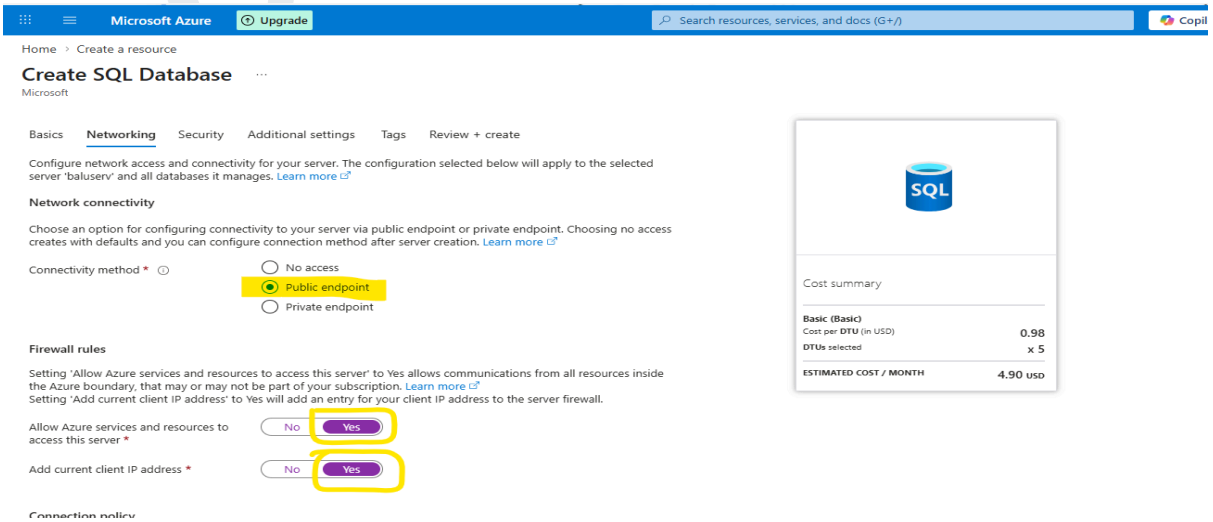
Network Tab

Network Connectivity Method: Public endpoint

Firewall Rules:

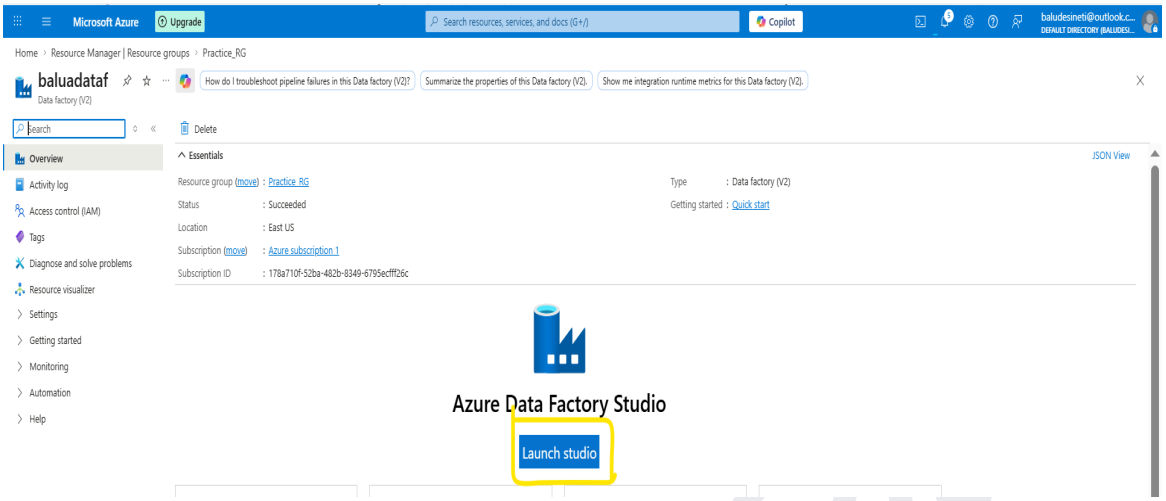
Allow Azure service and resources to access this service: Yes

Add Current Client IP Address: Yes



2.2 Azure Data Factory Setup:

Launching the Azure Factory Studio

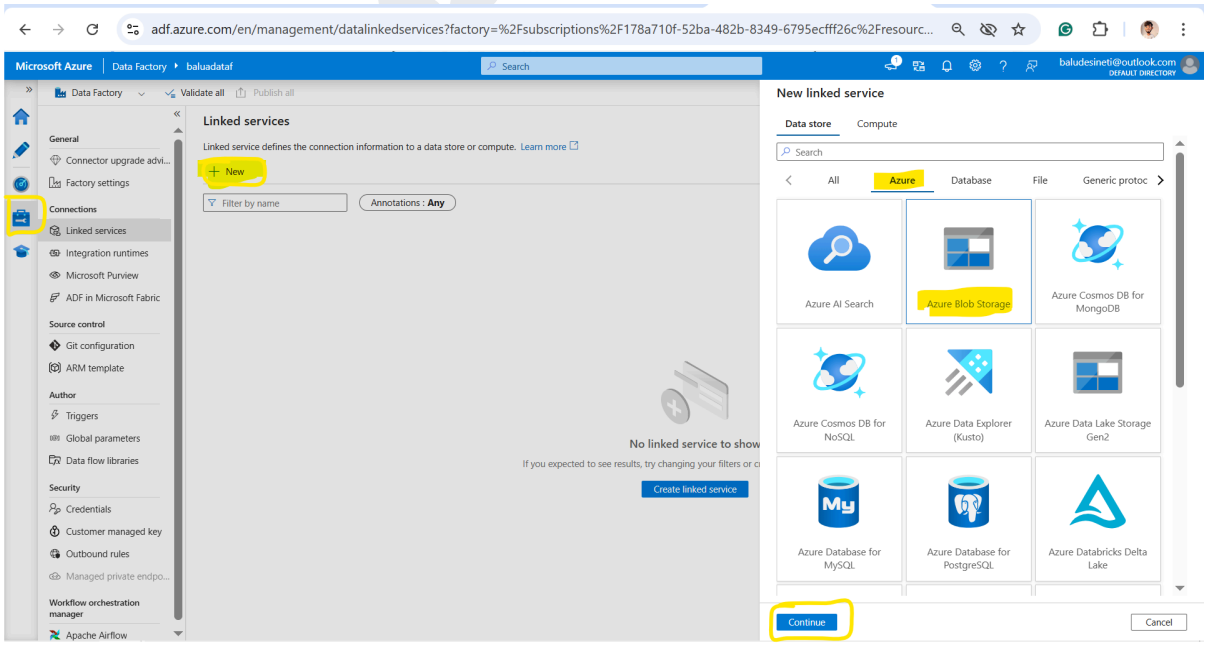


2.2.1 Linked Services:

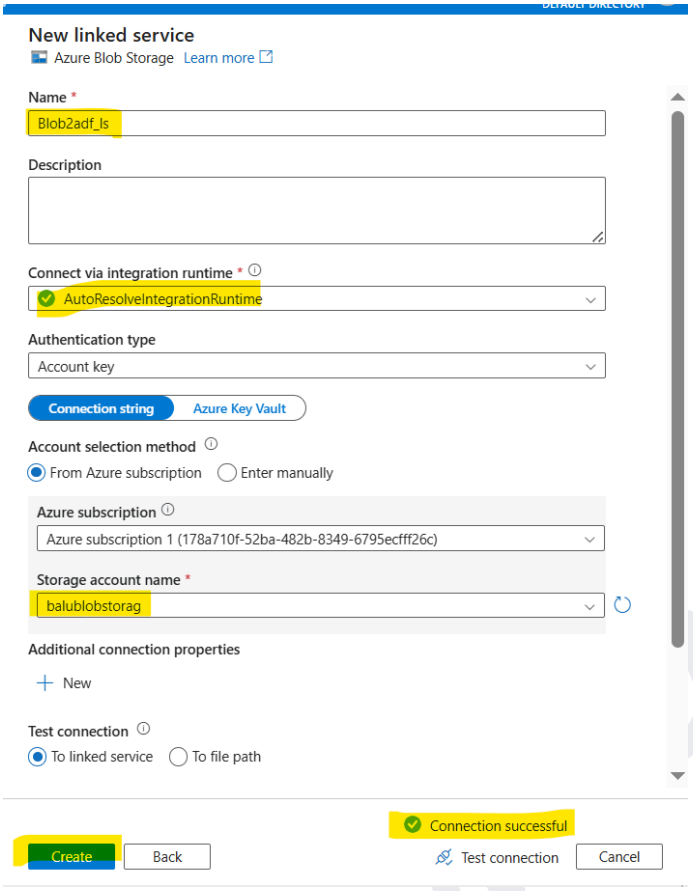
Linked Services define the connection information for external resources.

- **Azure Blob Storage:** Use Account Key or Service Principal for connection.

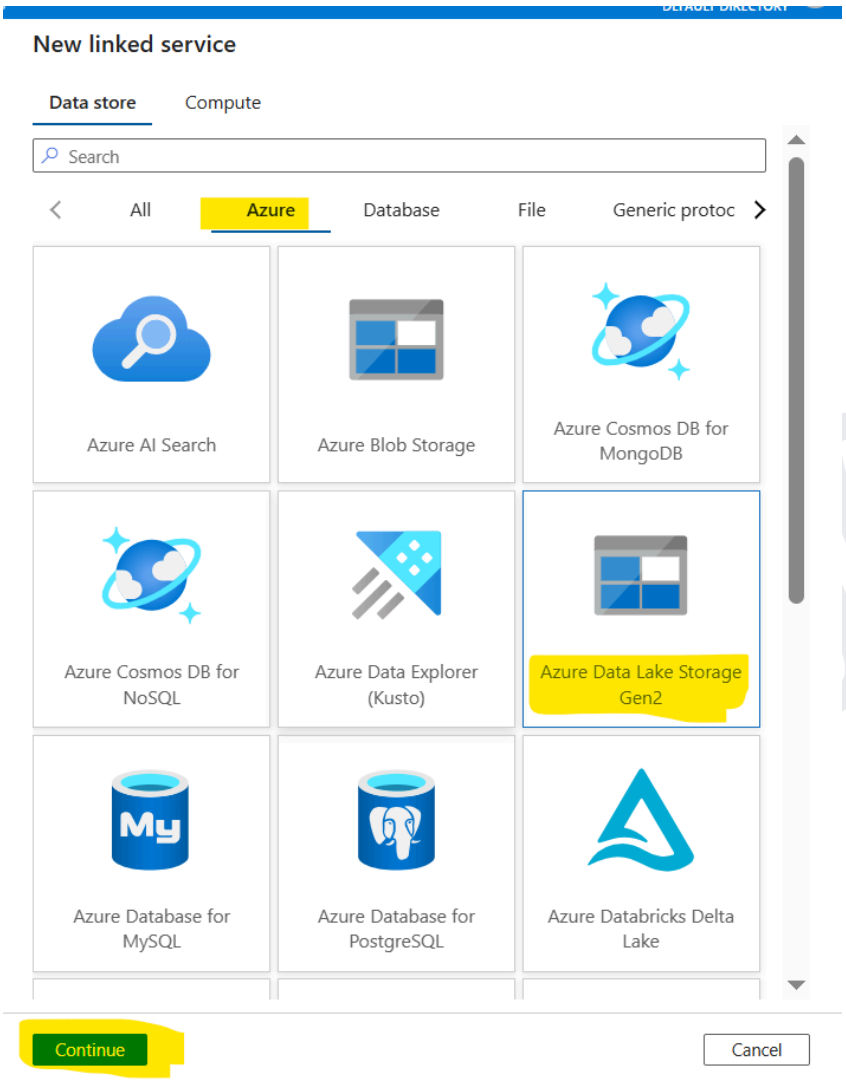
Navigate to the General → Linked Services → New → Azure → Azure Blob Storage → Create

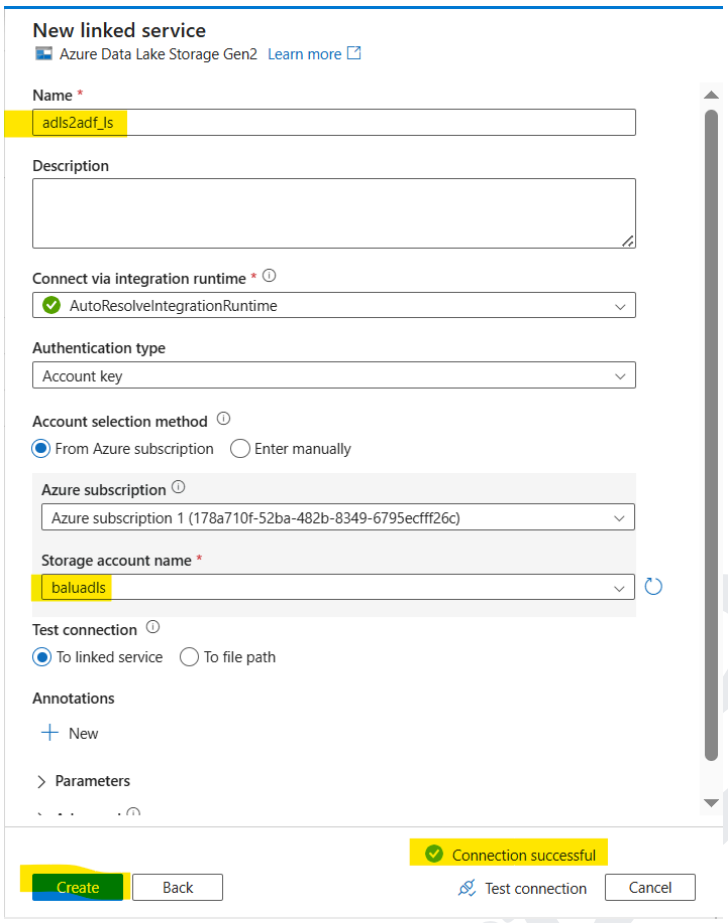


Name: **blob2adf_ls**
Connect via Integarion runtime: **AutoResolveIntegarionRuntime**
Storage Account Name: **balublobstorage**
Test connection: **click the Test Connection, which is successfully connected, then Create**



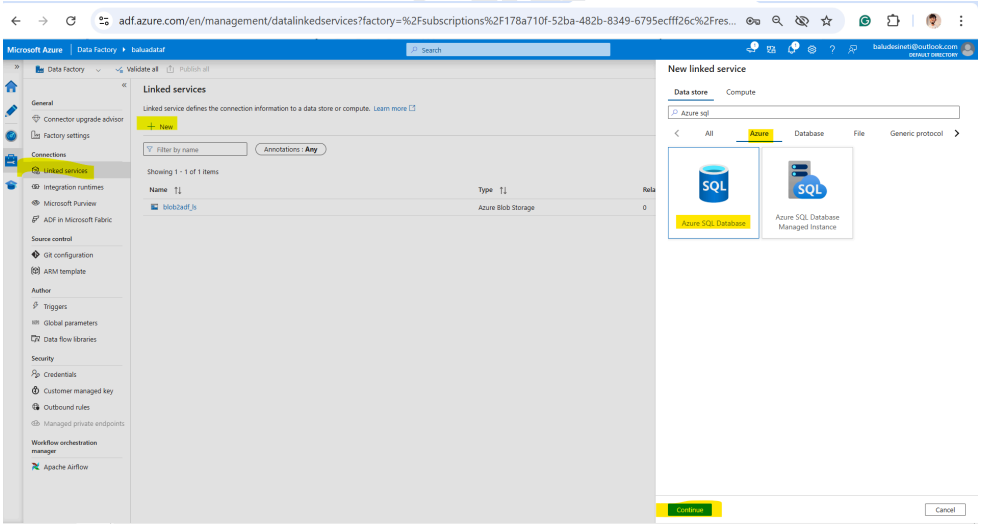
- **Azure Data Lake Storage Gen2:** Use Account Key, Service Principal, or Managed Identity.

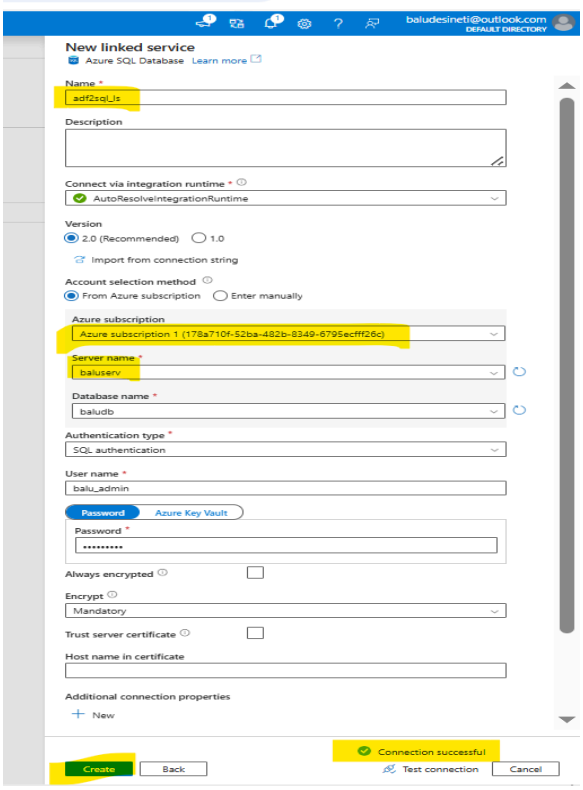

New linked service
Data store
Compute
Search
All
Azure
Database
File
Generic protoc
Azure AI Search
Azure Blob Storage
Azure Cosmos DB for MongoDB
Azure Cosmos DB for NoSQL
Azure Data Explorer (Kusto)
Azure Data Lake Storage Gen2
Azure Database for MySQL
Azure Database for PostgreSQL
Azure Databricks Delta Lake
Continue
Cancel



- **Azure SQL Database:** Use SQL Authentication or Azure Active Directory authentication.

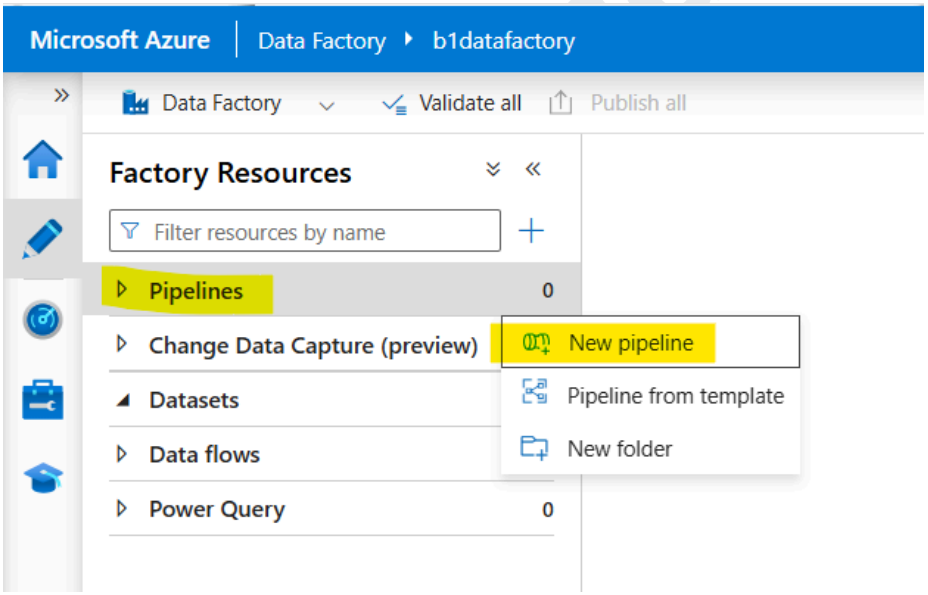
Click +New → Azure → Azure SQL Database – Create

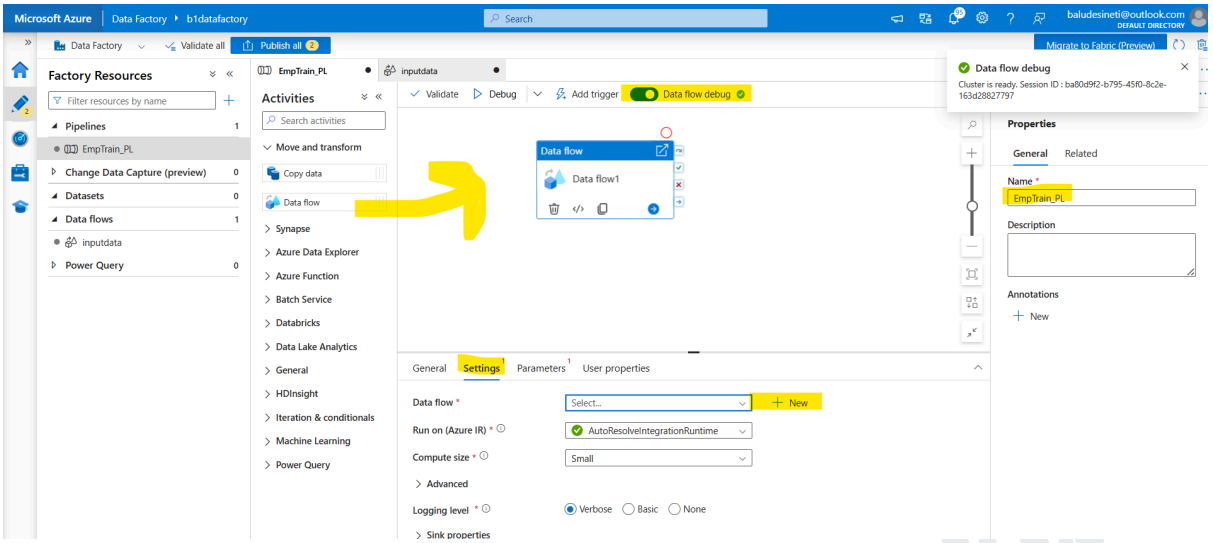


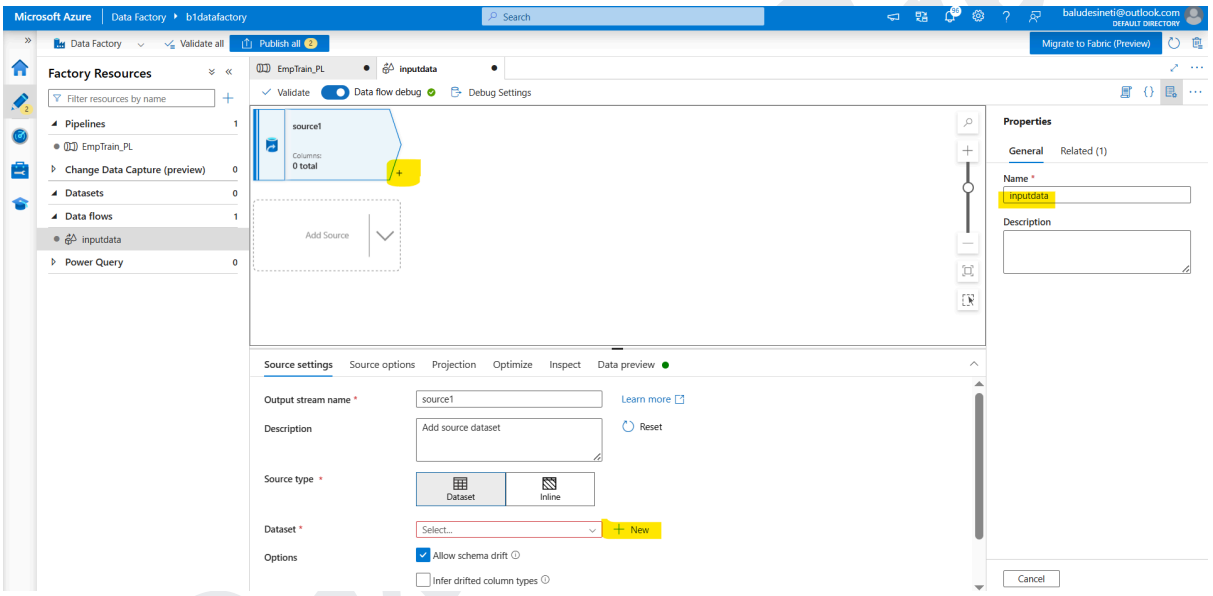


2.2 Create New Pipeline

Go to Author, Create New pipeline.




Microsoft Azure
Data Factory
b1datafactory
Search
baludesineti@outlook.com
DEFAULT DIRECTORY
Data Factory
Validate all
Publish all
Migrate to Fabric (Preview)
Factory Resources
Filter resources by name
Pipelines
EmpTrain_PL
Change Data Capture (preview)
Datasets
Data flows
inputdata
Power Query
EmpTrain_PL
inputdata
Activities
Search activities
Move and transform
Copy data
Data flow
Synapse
Azure Data Explorer
Azure Function
Batch Service
Databricks
Data Lake Analytics
General
HDInsight
Iteration & conditionals
Machine Learning
Power Query
Validate
Debug
Add trigger
Data flow debug
Data flow debug
Cluster is ready. Session ID : ba80d9f2-b795-45f0-8c2e-163d28827797
Data flow
Data flow1
Properties
General
Related
Name *
EmpTrain_PL
Description
Annotations
New
General
Settings
Parameters
User properties
Data flow *
Select...
New
Run on (Azure IR) *
AutoResolveIntegrationRuntime
Compute size *
Small
Advanced
Logging level *
Verbose
Basic
None
Sink properties


Microsoft Azure
Data Factory
b1datafactory
Search
baludesineti@outlook.com
DEFAULT DIRECTORY
Data Factory
Validate all
Publish all
Migrate to Fabric (Preview)
Factory Resources
Filter resources by name
Pipelines
EmpTrain_PL
Change Data Capture (preview)
Datasets
Data flows
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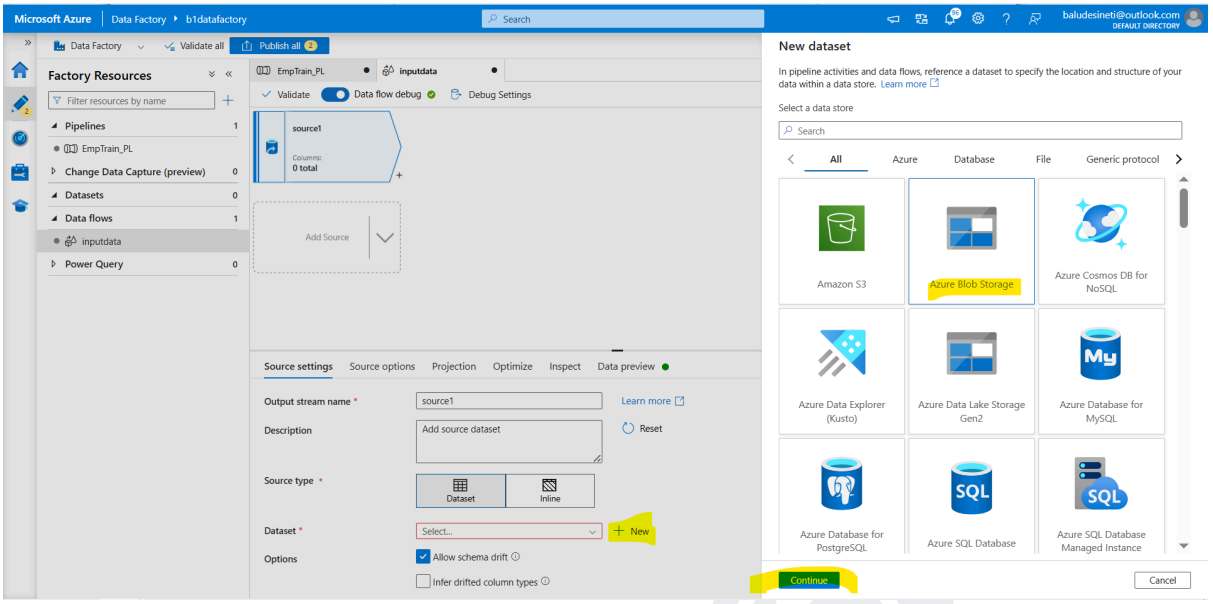
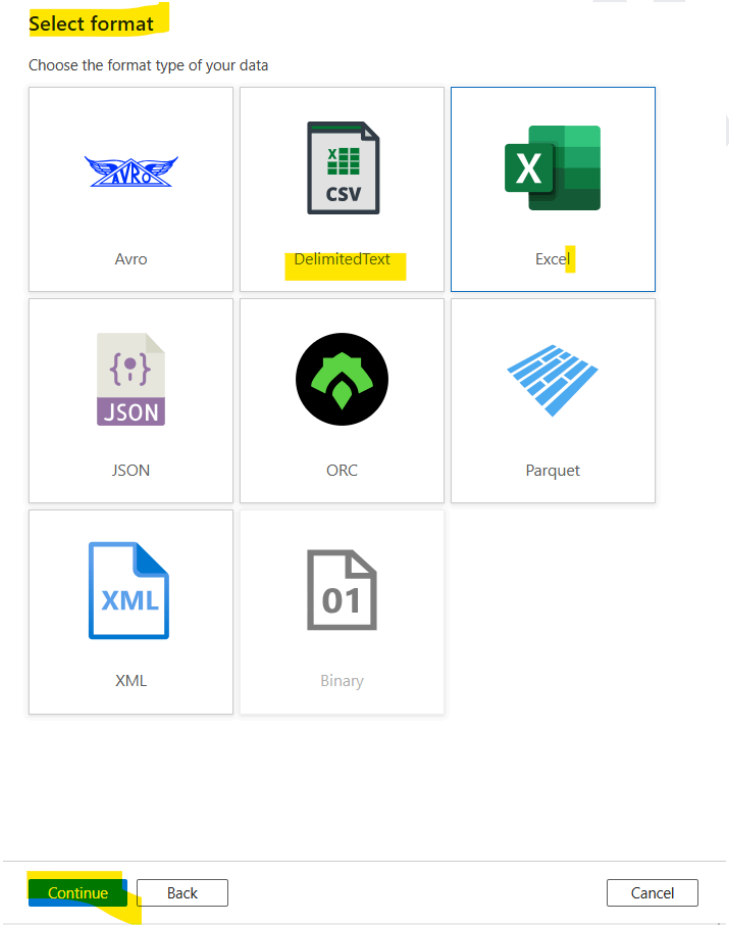
Power Query
EmpTrain_PL
inputdata
Validate
Data flow debug
Debug Settings
source1
Columns:
0 total
Add Source
Properties
General
Related (1)
Name *
inputdata
Description
Source settings
Source options
Projection
Optimize
Inspect
Data preview
Output stream name *
source1
Learn more
Description
Add source dataset
Reset
Source type *
Dataset
Inline
Dataset *
Select...
New
Options
Allow schema drift
Infer drifted column types
Cancel

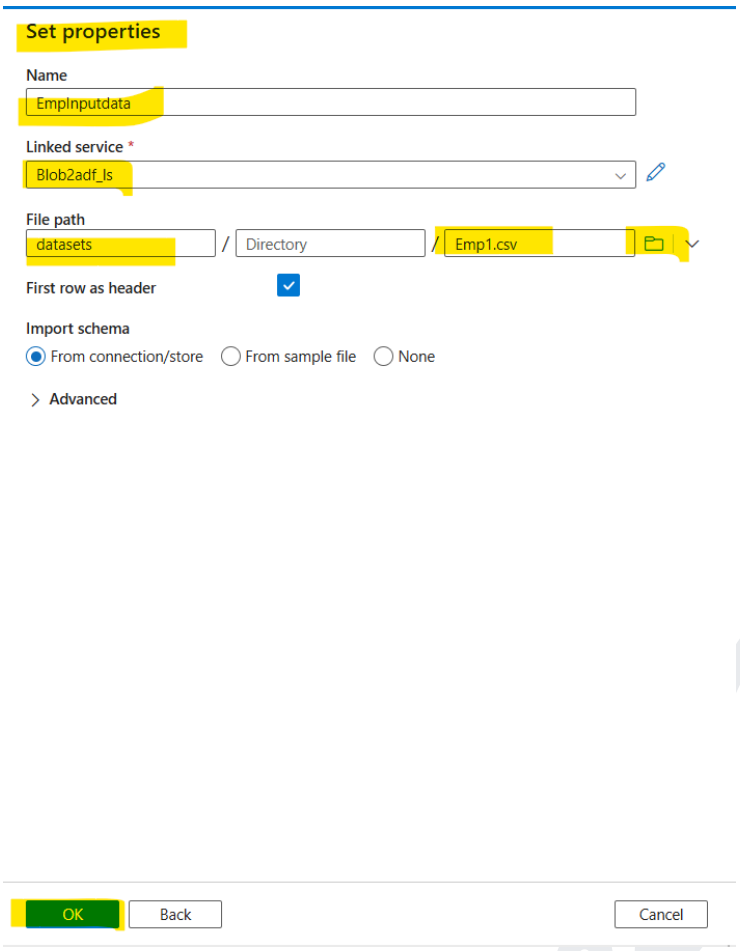
On the next page, it will ask which source file we have. We will select the source file located in the Azure blob and continue.



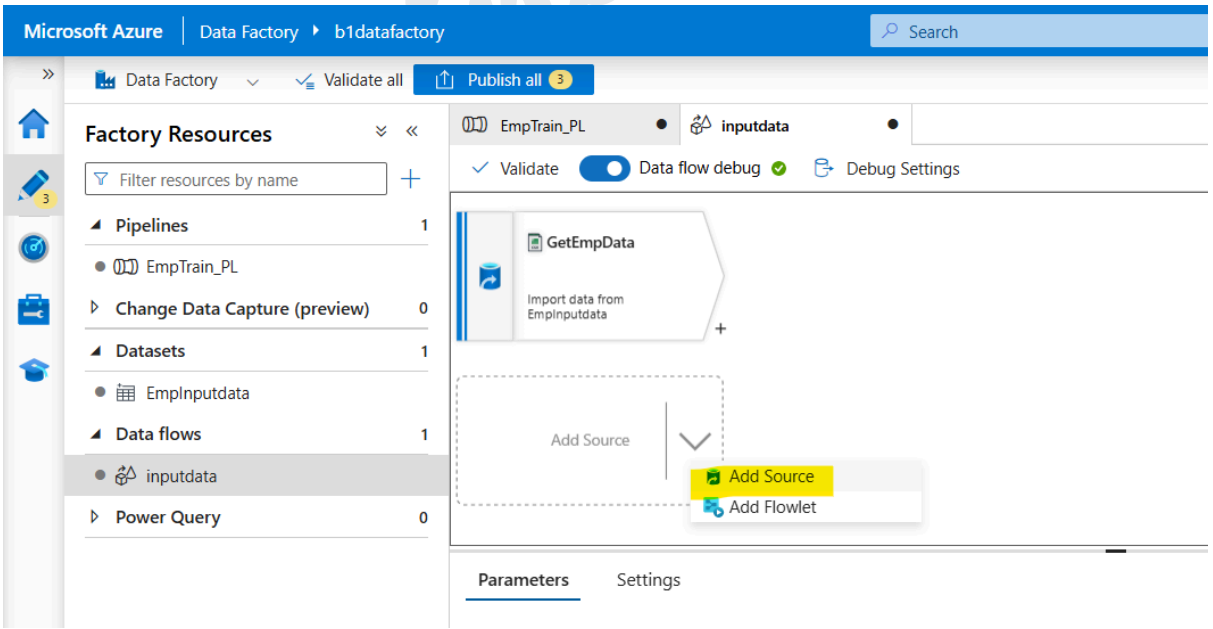
On the next Page, it will ask to select a format file, in this we have the CSV format and click on it and continue.



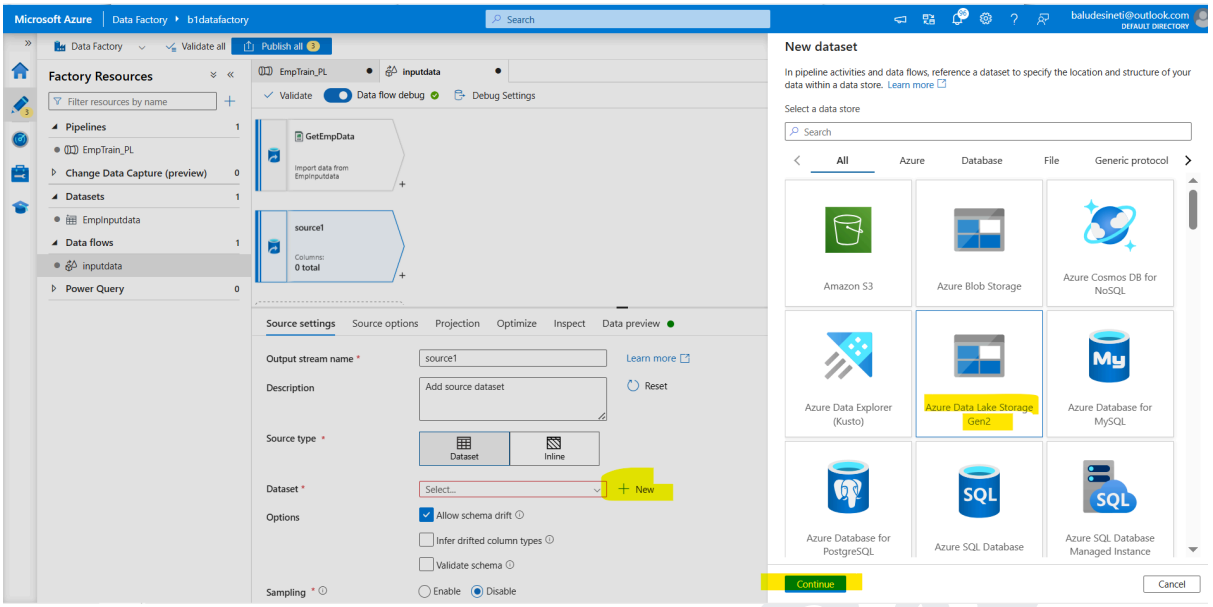
After that, it goes to the set Properties page, we need to give the link service "blob2adf_ls' and file path, and then click ok.



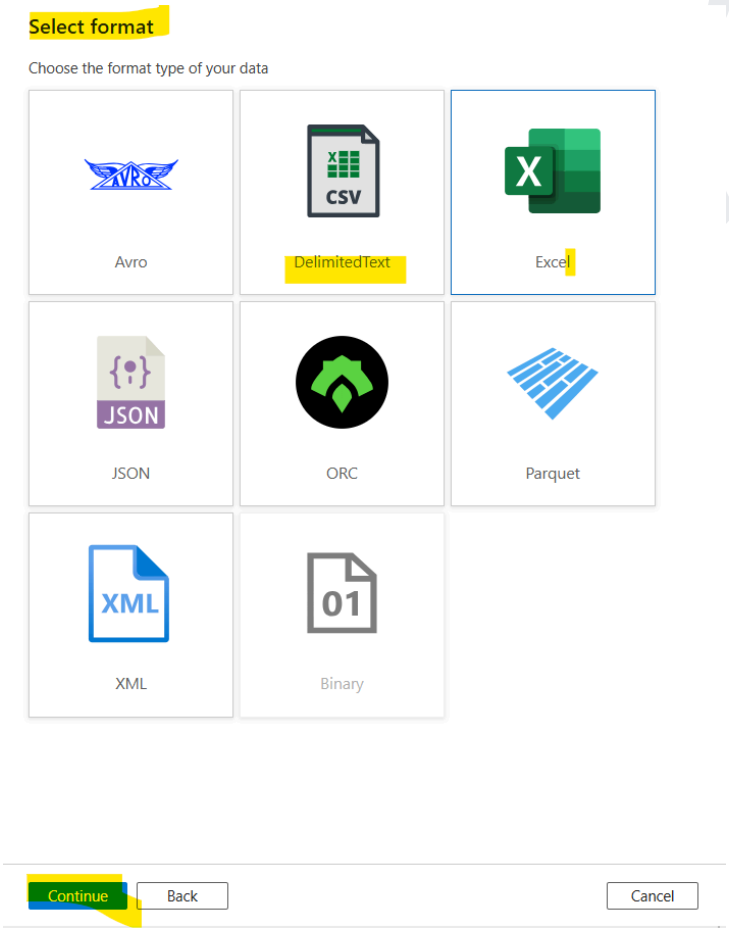
We need to add another source of Azure Data Lake Storage. We need to click the " Add source " button.



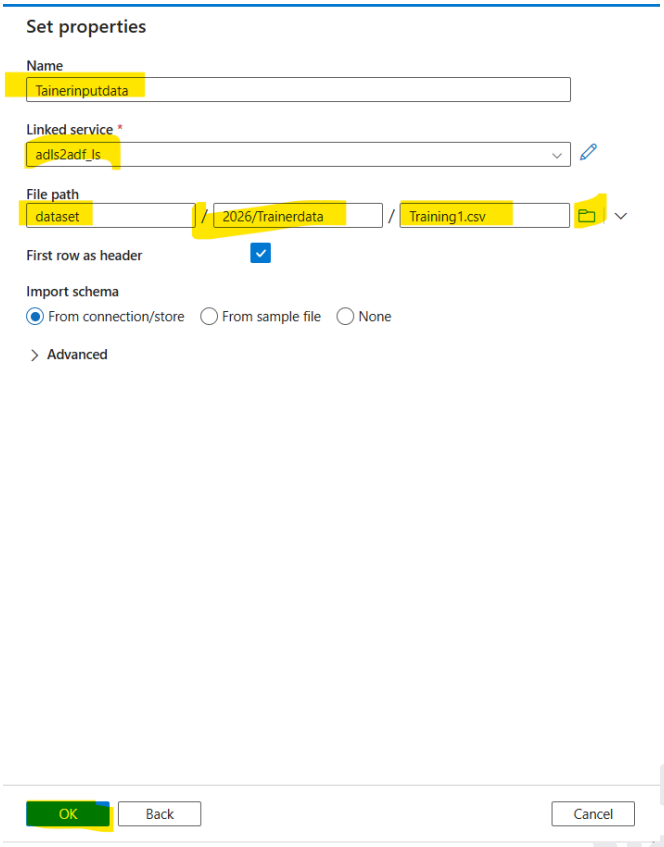
On the next page, it will ask which source file we have. We will select the source file located in the "Azure Data Lake Storage Gen 2" and continue.



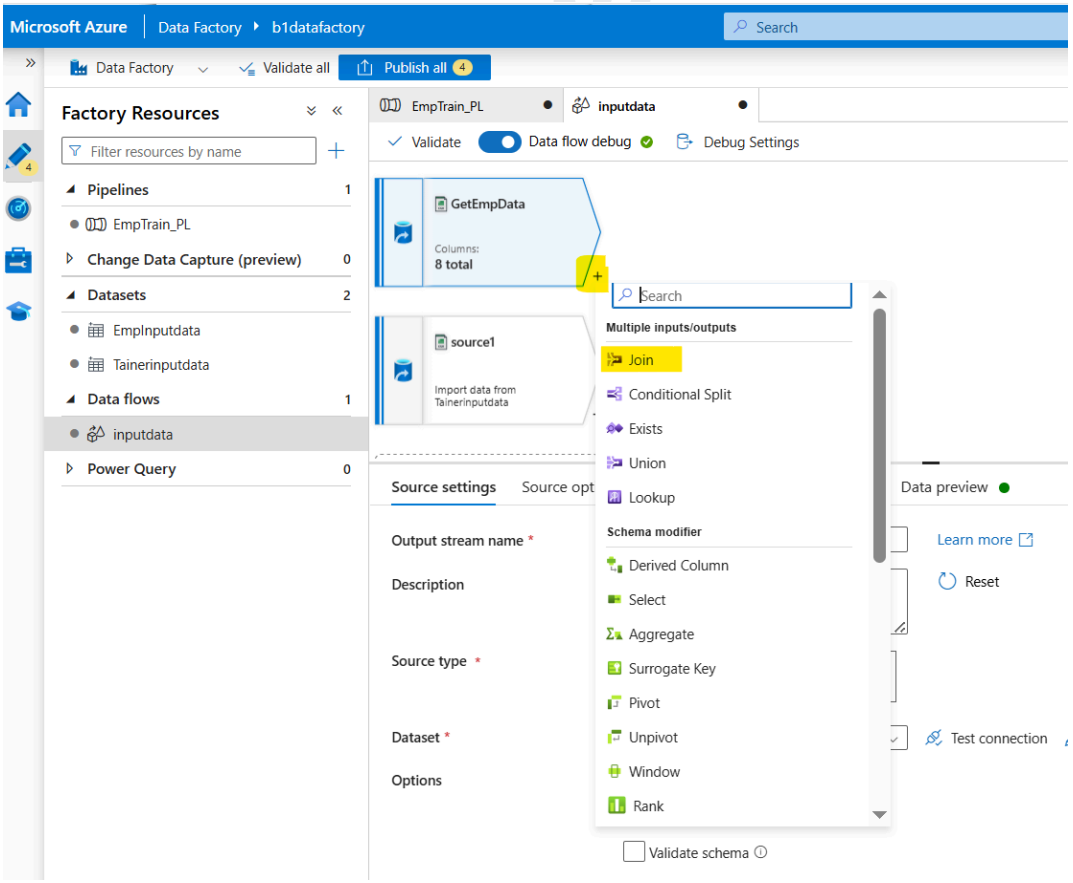
On the next Page, it will ask to select a format file, in this we have the CSV format and click on it and continue.



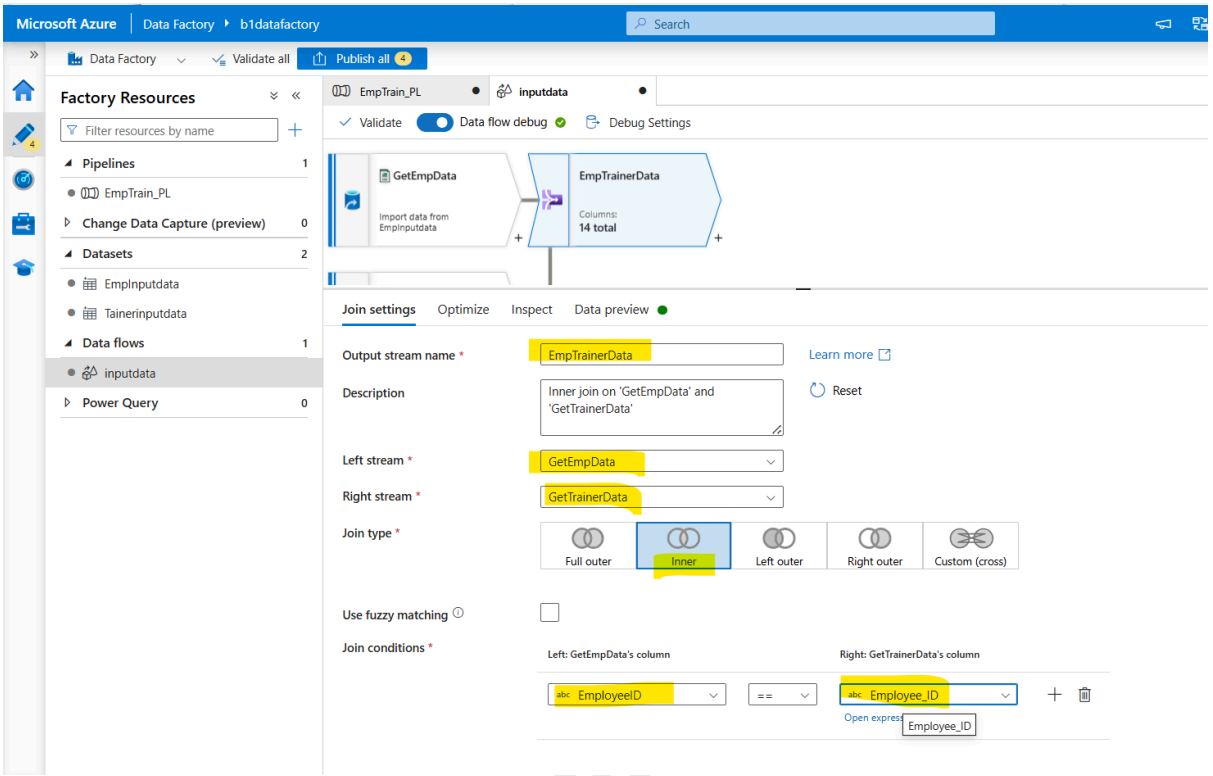
After that, it goes to the set Properties page, we need to give the link service "adls2adf_ls" and file path, and then click ok.



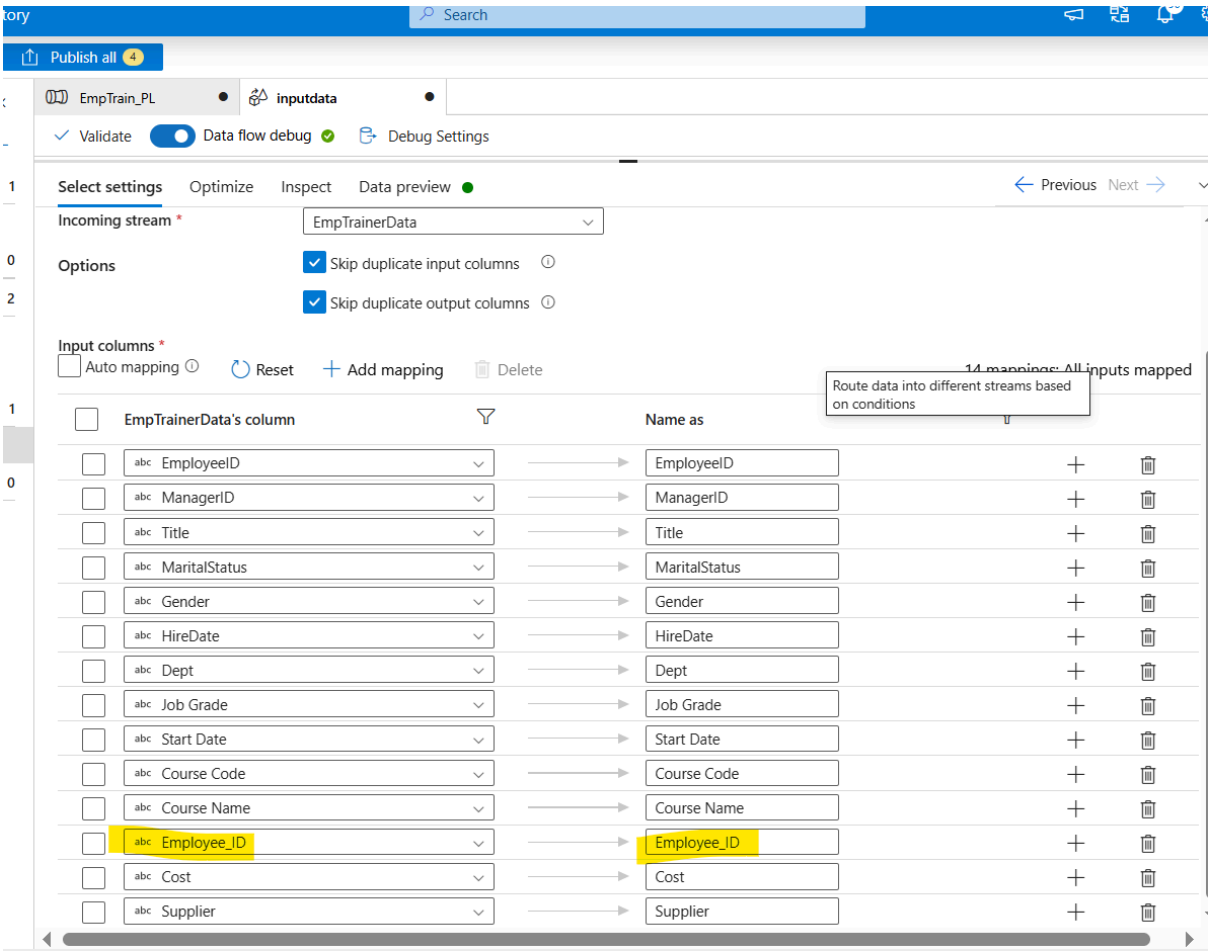
After all this, it will show the two sets, and we need to join both tables with the help of the "+" icon, it show the join we need to join 2tables



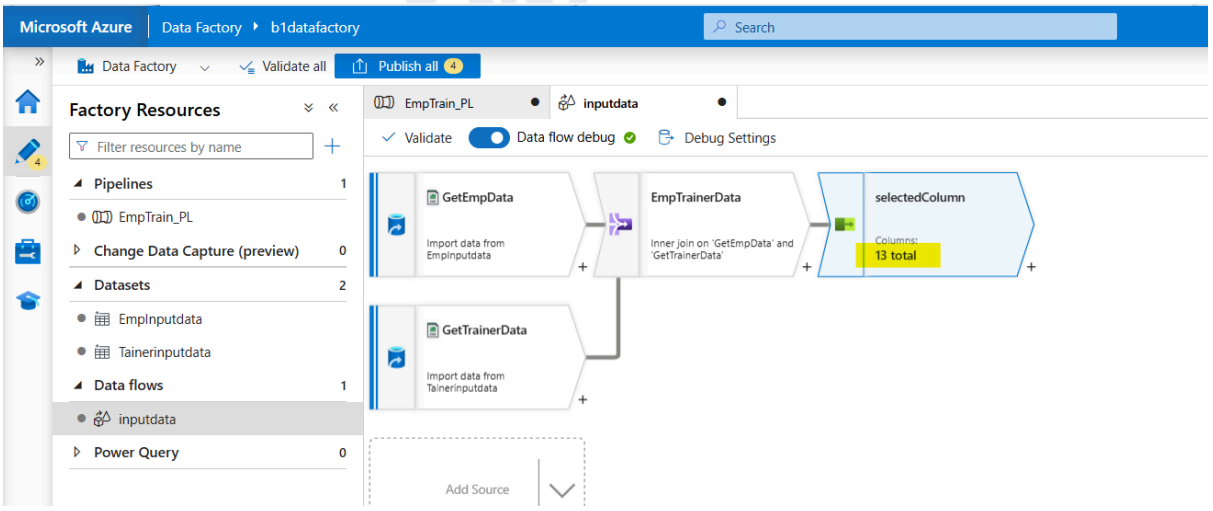
After we need to click the join table, we need to select the What type of join required, the join type is based on the common column of both table we need to select the column. After completing the join, it will show both table columns' count in the joined table.

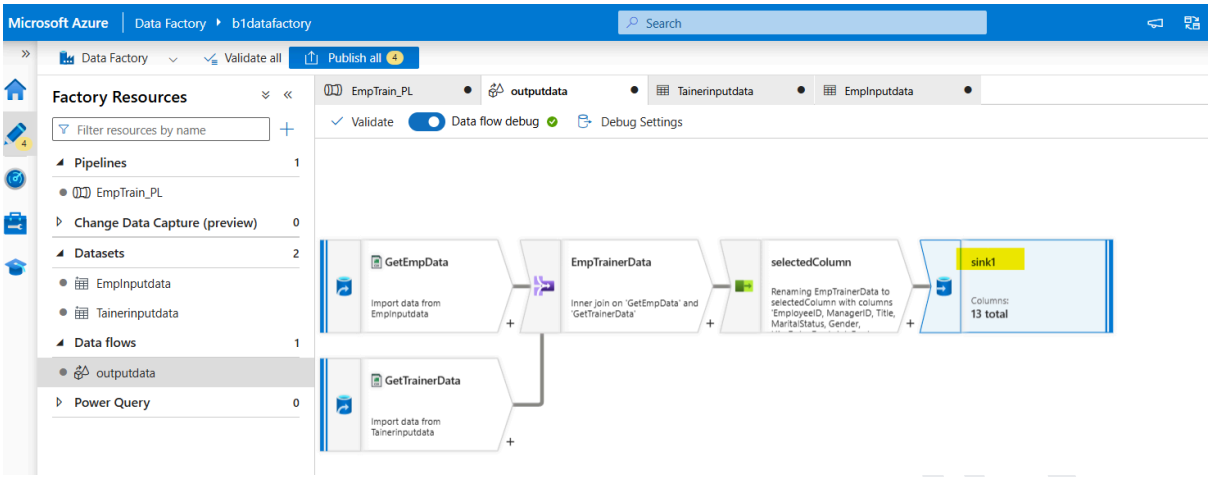


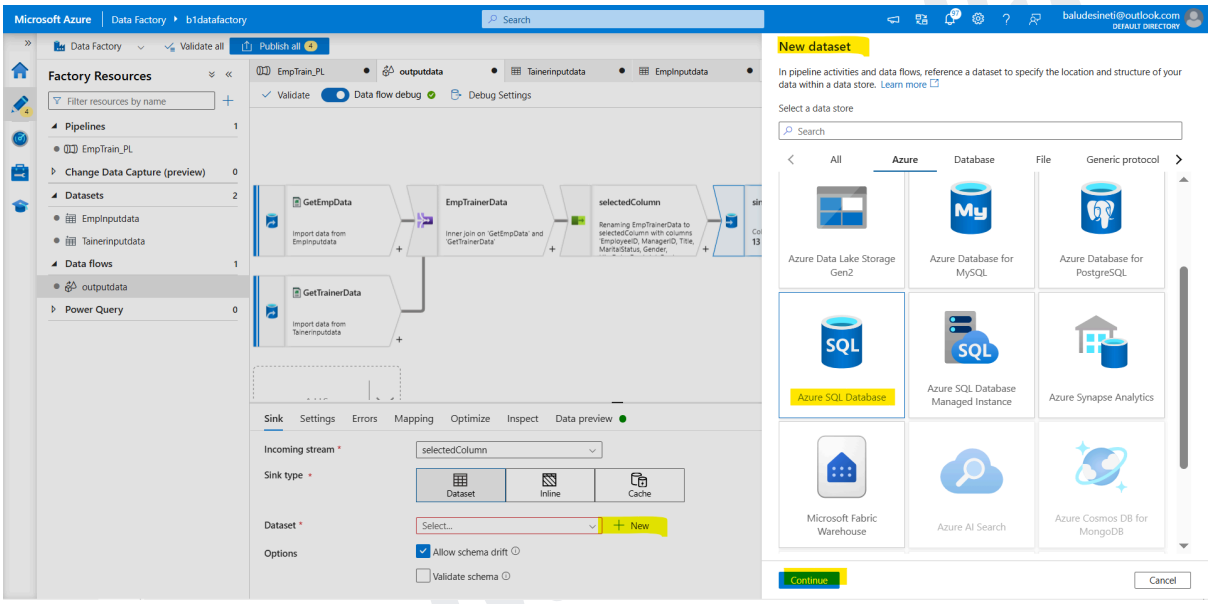
After joining is completed, we need to select the columns in both tables. We have the employee_ID, so we need to remove one column and clean the header name without spaces.



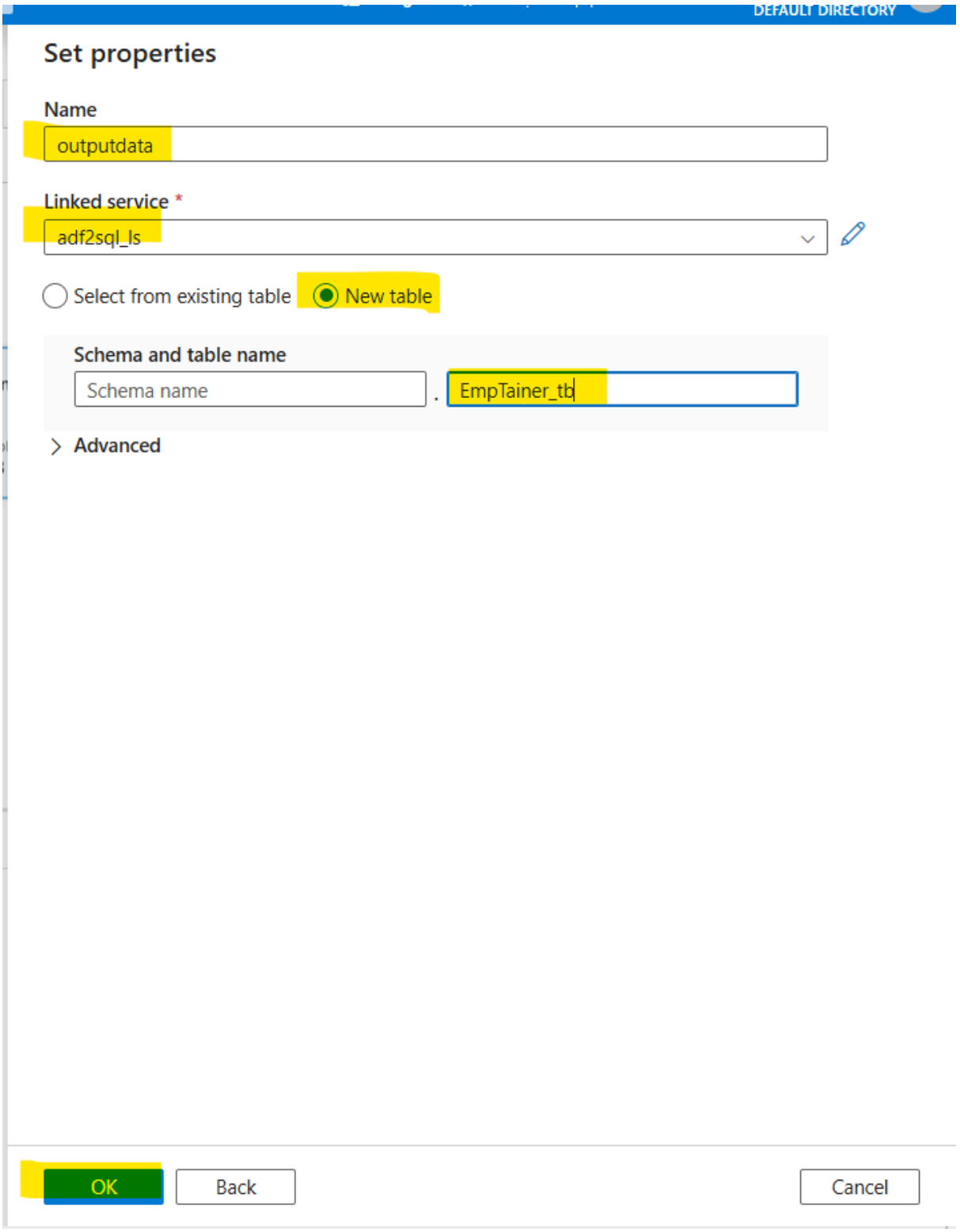
After cleaning the data set, it shows the 13columns. After completing all, we need to sink the data to the Azure SQL database.




Microsoft Azure
Data Factory
b1datafactory
Search
Data Factory
Validate all
Publish all 4
Factory Resources
Filter resources by name
Pipelines 1
EmpTrain_PL
Change Data Capture (preview) 0
Datasets 2
EmpInputdata
Tainerinputdata
Data flows 1
outputdata
Power Query 0
EmpTrain_PL
outputdata
Tainerinputdata
EmpInputdata
Validate
Data flow debug
Debug Settings
GetEmpData
Import data from EmpInputdata
EmpTrainerData
Inner join on 'GetEmpData' and 'GetTrainerData'
selectedColumn
Renaming EmpTrainerData to selectedColumn with columns 'EmployeeID, ManagerID, Title, MaritaStatus, Gender,
sink1
Columns:
13 total
GetTrainerData
Import data from Tainerinputdata


Microsoft Azure
Data Factory
b1datafactory
Search
baludesineti@outlook.com
DEFAULT DIRECTORY
Data Factory
Validate all
Publish all 4
Factory Resources
Filter resources by name
Pipelines 1
EmpTrain_PL
Change Data Capture (preview) 0
Datasets 2
EmpInputdata
Tainerinputdata
Data flows 1
outputdata
Power Query 0
Validate
Data flow debug
Debug Settings
GetEmpData
EmpTrainerData
selectedColumn
GetTrainerData
Sink
Settings
Errors
Mapping
Optimize
Inspect
Data preview
Incoming stream *
selectedColumn
Sink type *
Dataset
Inline
Cache
Dataset *
Select...
New
Options
Allow schema drift
Validate schema
New dataset
In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more
Select a data store
Search
All
Azure
Database
File
Generic protocol
Azure Data Lake Storage Gen2
Azure Database for MySQL
Azure Database for PostgreSQL
Azure SQL Database
Azure SQL Database Managed Instance
Azure Synapse Analytics
Microsoft Fabric Warehouse
Azure AI Search
Azure Cosmos DB for MongoDB
Continue
Cancel

After selected azure sql database, it will go to the Set Properties. We need to give the name and linked service "adf2sql_ls" and the new table name EmpTrainer_tb, and clickon ok button.



After completing all, we need to save our work so we can publish it. After publishing all, we need to execute the pipeline,ine so click on the tiggernow button.

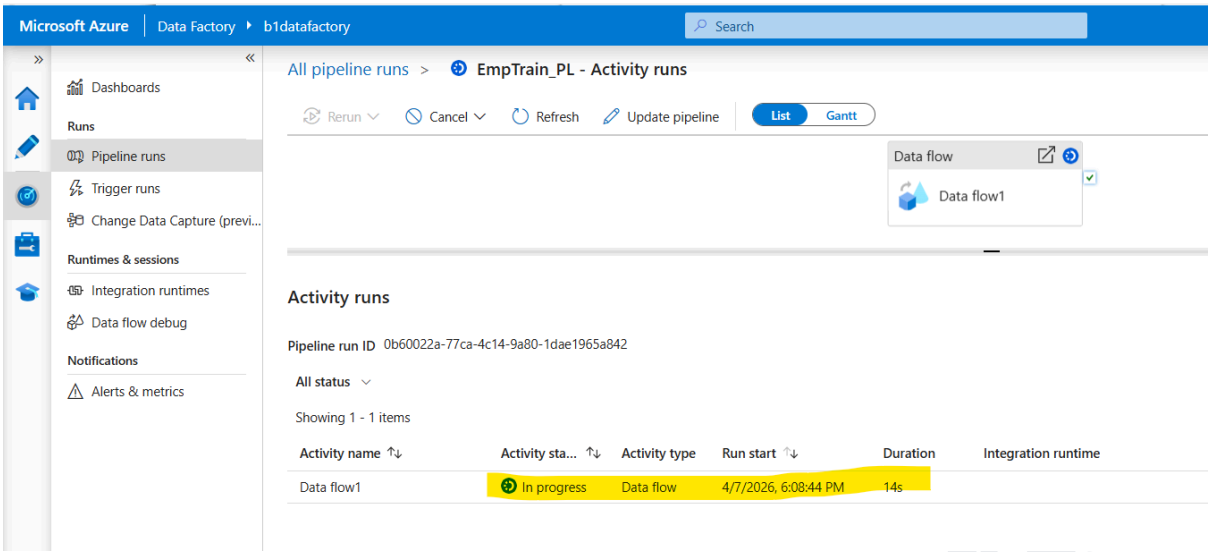
Publish all

You are about to publish all pending changes to the live environment. Learn more

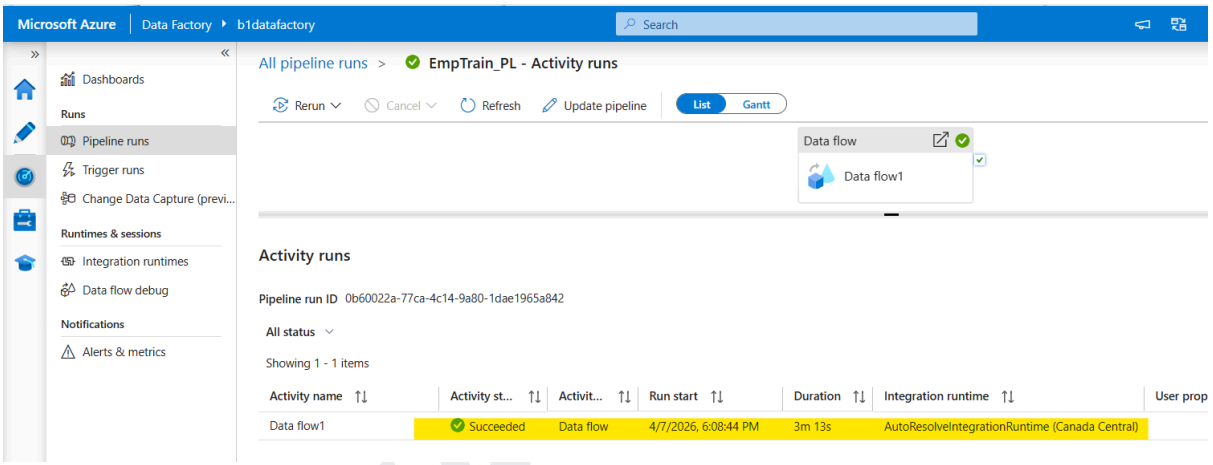
Pending changes (5)

NAME	CHANGE	EXISTING
Pipelines		
EmpTrain_PL	(New)	-
Datasets		
EmpInputdata	(New)	-
Tainerinputdata	(New)	-
outputdata	(New)	-
Data flows		
EmpTainer_df	(New)	-

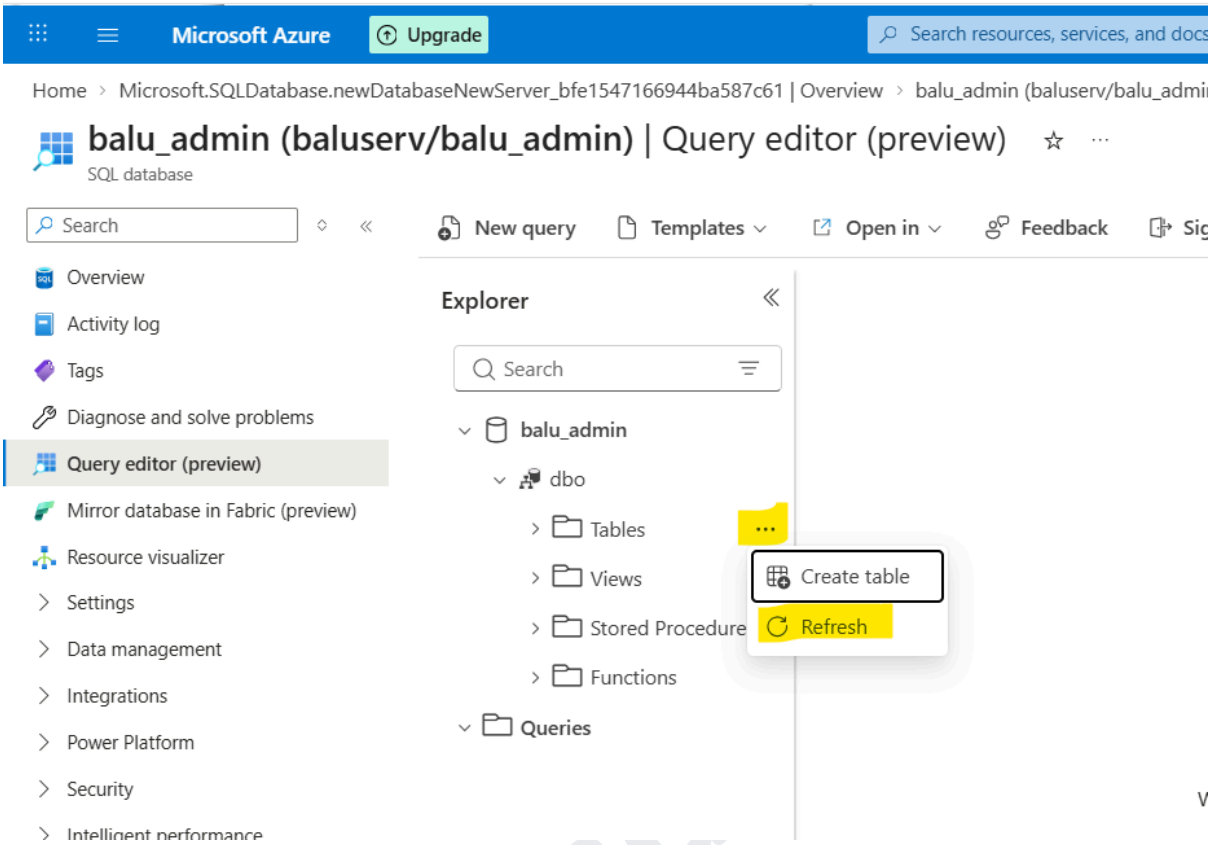
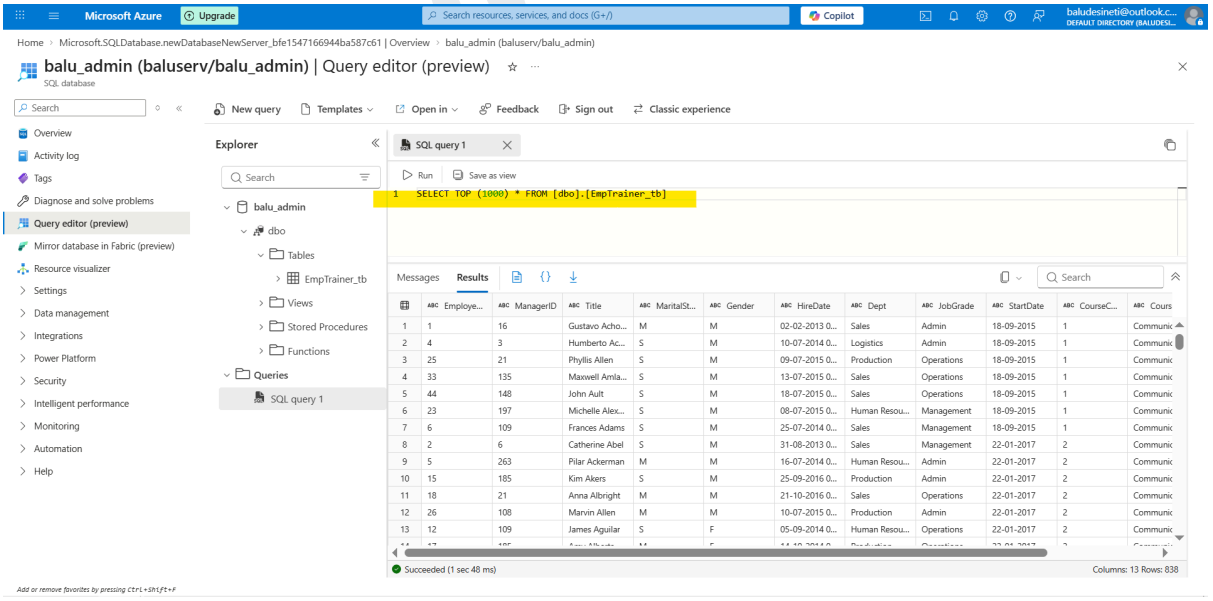
Publish Cancel



Successfully Pipeline executed, now we need to the azure sql database to chcke the data availability on the Azure SQL database.



After entering the Azure SQL database, we need to refresh the database, and then it will show the data.



Successfully Completed the Azure Blob + Azure Data Lake Store to Azure Sql database (Using Dataflow)